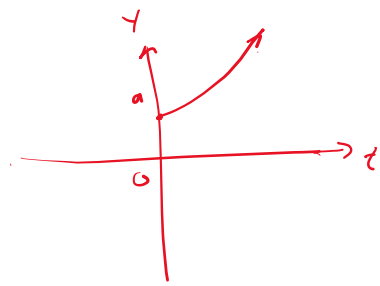
Exponential Applications 1

1. **Exponential growth and decay**



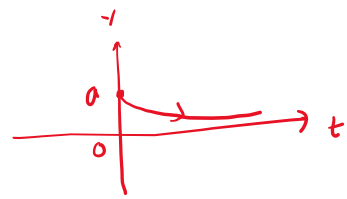
1. Exponential Growth

$$y = a(1+r)^t$$

When to use it: When a value is increasing by a percentage over time.

- **Common Examples:** Population growth, home appreciation, or a single investment that you leave alone to grow (not adding monthly payments).
- **The "Growth Factor":** The $(1+r)$ part is always greater than 1.

2. Exponential Decay



$$y = a(1-r)^t$$

When to use it: When a value is decreasing by a percentage over time.

- **Common Examples:** The value of a car as it gets older (depreciation), radioactive decay, or the caffeine levels in your body after drinking a cup of coffee.
- **The "Decay Factor":** The $(1-r)$ part is always less than 1.

Feature	Growth Formula	Decay Formula
Operator	Plus $(+)$	Minus $(-)$
Trend	Value goes **Up**	Value goes **Down**
Real-world Example	A $\$1,000$ antique that gains 10% value each year.	A $\$1,000$ laptop that loses 10% value each year.

Example 1: The number of bachelor's business degrees (in thousands) conferred in the United States can be modeled by an exponential function. In 2010, 279,000 business degrees were conferred, and the number of degrees has risen 2% each year. exponential growth

a. Write an equation to model the number of business degrees conferred.

$$y = a \cdot (1 + r)^t$$

$r = 2\%$

$a = 279,000$

t : the number of year since 2010

[In 2010, $t = 0$]

y : the number of business degrees conferred

$$\Rightarrow \quad y = 279,000 * (1.02)^t$$

b. Graph the equation

c. What was the first year in which 475,000 business degrees were conferred?

$$y = 475,000$$

$$\Leftrightarrow \quad 279,000 * (1.02)^t = 475,000$$

$$\Leftrightarrow \quad (1.02)^t = \frac{475,000}{279,000} \approx 1.703$$

$$\Leftrightarrow \quad (1.02)^t = 1.703$$

$$\Leftrightarrow \quad \log 1.02^t = \log 1.703$$

$$\Leftrightarrow \quad t * \log 1.02 = \log 1.703$$

$$\Leftrightarrow \quad t = \frac{\log 1.703}{\log 1.02} \approx 26.885$$

found up to $t = 27$

The first year when $Y = 475{,}000$ is $2010 + 27 = \boxed{2037}$

d. Use the model to estimate how many degrees will be conferred in 2024.

In 2024, $t = 2024 - 2010 = 14$

$$Y = 279{,}000 * (1.02)^{14} = 368{,}134.57$$

$$\approx 368135$$

Example 2: A piece of machinery is purchased for $40,000. The business estimates that its value will decrease 43% each year.

a. Write an equation to model the value of the machinery after *t* years.

$y = a \cdot (1 - r)^t$

b. Graph the equation

c. What is the value of the machinery after 2 years?

d. When will the machinery be worth $5,000?

3. Interest

① Suppose we have P amount of money.

② Deposit it to a bank to grow

- Grow linearly | simple interest
- grow exponentially | compound interest
  - multiple compound
  - continuous compound

Example: $P = 1000$

Interest rate $= 10\%$

	simple interest	Compound Interest
Year 0	$P = 1000$	$P = 1000$
Year 1	$P = 1000 + 1000 * 10\%$ $= 1000(1 + .1)$ $= 1100$	$P = 1000 + 1000 * 10\%$ $= 1000 * (1 + .1)$ $= 1100$
Year 2	$P = 1100 + 1000 * 10\%$ $P = \underline{1200}$	$P = 1100 + 1100 * 10\%$ $P = 1100 + 110 = \underline{1210}$

year t :	$P = 100 \cdot (1 + .1 \cdot t)$	$P = 100\,(1 + .1)^t$

a. Multiple Compound Interest

The formula for compound interest is:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

Understanding the Variables

- A: The final amount of money accumulated after t years, including interest.
- P: The principal amount (the initial money you start with).
- r: The annual interest rate (expressed as a decimal; for example, 5% becomes 0.05).
- n: The number of times that interest is compounded per year (e.g., 12 for monthly, 365 for daily).
- t: The number of years the money is invested or borrowed for.

Example 5: You place $5,000 in a certificate of deposit (CD) with a 3% interest rate for 2 years. Find the value of the CD at the different compounding periods. Then find the interest earned.

a. Annually

$P = 5000$ $\quad n = 1$

$r = 3\%$

$t = 2$

$$A = P \cdot \left(1 + \frac{r}{n}\right)^{nt}$$

$$A = 5000\left(1 + \frac{.03}{1}\right)^{1 \cdot 2} = 5304.5$$

Interest earned : $5304.5 - 5000 = 304.5$

b. Semiannually (2 times a year)

$n = 2$:

$$\rightarrow \quad A = 5000\left(1 + \frac{.03}{2}\right)^{2 \cdot 2} = 5306.81$$

c. Quarterly (4 times a year)

$n = 4$

$$A = 5000\left(1 + \frac{.03}{4}\right)^{4 \cdot 2} = 5307.99$$

d. Monthly

$$n = 12 \rightarrow A = 5000\left(1 + \frac{.03}{12}\right)^{12 \cdot 2} = 5308.78$$

e. Weekly

$n = 52$

f. Daily

$n = 365$

Example 6: What amount deposited today at 2.8% compounded monthly will amount to $6000 in 5 years?

$$A = P \cdot \left(1 + \frac{r}{n}\right)^{nt}$$

Example 7: You want to buy your first car upon graduation. You would like to have $4,000 as a down payment. Find the interest rate, with quarterly compounding, that will allow you to grow $2,500 to $4,000 in four years.

Example 8: You have $2000 from high school graduation. You place it in a high yield savings account with 4.35% interest compounded monthly and make no additional deposits. How long will it take for your money to double?

b. Continuously compounding interest

The formula for continuous compounding is:

$$A = Pe^{rt}$$

Understanding the Variables

- A: The final amount.
- P: The principal amount.
- e: The base of the natural logarithm (Euler's number, ≈ 2.71828).
- r: The annual interest rate (in decimal form).
- t: The time in years.

Example 9: If you invest $8,000 at an annual interest rate of 1.9% compounded continuously for 3 years, how much will you have at the end of the three years? How much interest did you earn?

Example 10: You are saving up for a down payment on your first home. You would like to have $20,000. You are giving yourself 10 years to get to this amount. You have found an account that offers continuous compounding. What does the interest rate need to be to grow $10,000 to $20,000 over 10 years?

Example 11: Following the birth of their granddaughter, two grandparents want to make an initial investment that will grow to $10,000 by the child's 20th birthday. Interest is compounded continuously at 6%. What should the initial investment be?

Example 12: How long will it take a company's $50,000 investment to triple if it is invested in an account offering continuous compounding at 4.9%?